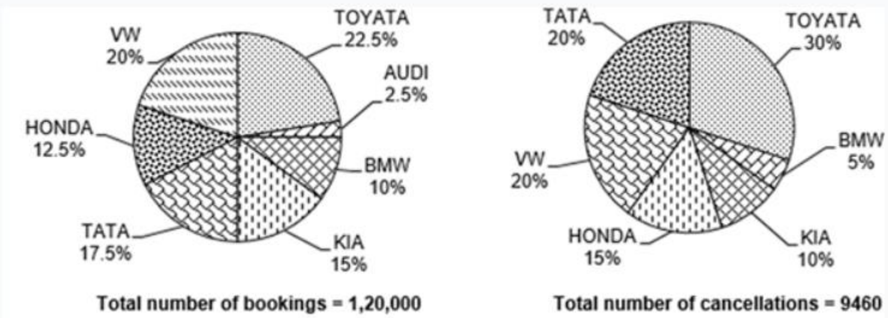
DI – PIE CHARTS

Directions for questions 1 to 5: These questions are based on the pie charts given below which provide the information about

a. the number of bookings made for cars of various companies in 2019 and

b. the breakup of cancellations for the bookings made in 2019, both in city x.



Number of cars delivered = Number of bookings made – Number of cancellations

What is the ratio of the total number of cars delivered by the companies TATA, BMW and AUDI to the total number of bookings made for the cars of the companies TOYOTA, BMW and VW?

a) $\frac{961}{1600}$

b) $\frac{941}{1600}$

c) $\frac{961}{1800}$ (Correct Answer - ✅)

d) $\frac{943}{1600}$

Solution:

Company	Number of bookings made	Number of Cancellations made	Number of cars delivered
AUDI	3,000	0	3,000
BMW	12,000	473	11,527
KIA	18,000	946	17,054
TATA	21,000	1,892	19,108
HONDA	15,000	1,419	13,581
VW	24,000	1,892	22,108
TOYOTA	27,000	2,838	24,162

The required ratio = $\frac{3,000 + 11,527 + 19,108}{27,000 + 24,000 + 12000}$

$= \frac{33,635}{63,000} = \frac{961}{1800}$ Choice (C)

The number of cars delivered by KIA is approximately what percentage of the number of bookings made for VW?

a) 71% (Your answer - ✓)

b) 72%

c) 73%

d) 74%

Solution:

Company	Number of bookings made	Number of Cancellations made	Number of cars delivered
AUDI	3,000	0	3,000
BMW	12,000	473	11,527
KIA	18,000	946	17,054
TATA	21,000	1,892	19,108
HONDA	15,000	1,419	13,581
VW	24,000	1,892	22,108
TOYOTA	27,000	2,838	24,162

The required percentage = $\frac{17,054}{24,000} \times 100$

$\cong \frac{17}{24} \times 100 \cong 71\%$ Choice (A)

The ratio of the number of cars delivered to the number of cancellations made is the least for which of the following companies?

- a) TATA (Your answer - ✗)
- b) HONDA (Correct Answer - ✓)
- c) VW
- d) KIA

Solution:

Company	Number of bookings made	Number of Cancellations made	Number of cars delivered
AUDI	3,000	0	3,000
BMW	12,000	473	11,527
KIA	18,000	946	17,054
TATA	21,000	1,892	19,108
HONDA	15,000	1,419	13,581
VW	24,000	1,892	22,108
TOYOTA	27,000	2,838	24,162

Except for HONDA, for all the other companies (given in the options) the ratio of the number of cars delivered to the number of cancellations made is greater than 10.

Choice (B)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

What is the difference between the number of cars delivered by TATA and the number of cancellations made for TATA cars?

Solution:

Company	Number of bookings made	Number of Cancellations made	Number of cars delivered
AUDI	3,000	0	3,000
BMW	12,000	473	11,527
KIA	18,000	946	17,054
TATA	21,000	1,892	19,108
HONDA	15,000	1,419	13,581
VW	24,000	1,892	22,108
TOYOTA	27,000	2,838	24,162

The required difference = 19,108 – 1,892 = 17,216 Ans : (17216)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

What is the total number of cars delivered by all the given companies?

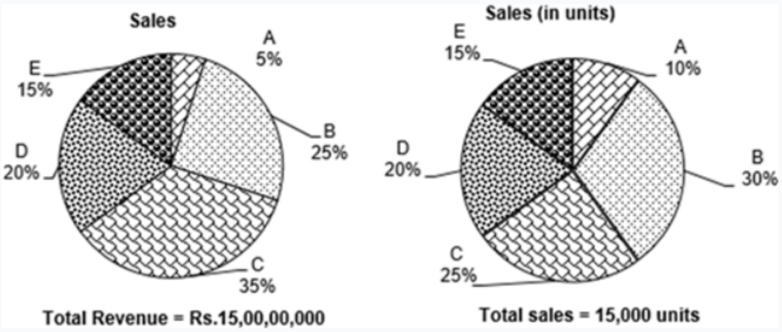
Solution:

Company	Number of bookings made	Number of Cancellations made	Number of cars delivered
AUDI	3,000	0	3,000
BMW	12,000	473	11,527
KIA	18,000	946	17,054
TATA	21,000	1,892	19,108
HONDA	15,000	1,419	13,581
VW	24,000	1,892	22,108
TOYOTA	27,000	2,838	24,162

The total number of cars delivered = 3,000 + 11,527 + 17,054 + 19,108 + 13,581 + 22,108 + 24,162 = 1, 10,540 Ans : (110540)

Directions for questions 6 to 10: Answer the following questions based on the information given below.

The following pie charts provide the information about the sales (in Rs.) and sales (in quantity) of five different products – A, B, C, D and E – of company XYZ.



Cost of manufacturing of products A, B, C, D and E are Rs.4100, Rs.4000, Rs.3800, Rs.3400 and Rs.3200 respectively.

Profit = Selling price – Cost price

Profit percentage =

$$\frac{\text{Profit}}{\text{Cost price}}$$

× 100

All the products manufactured are sold.

For how many products is the selling price per unit more than the average selling price of all the units sold?

- a) 1
- b) 2
- c) 3
- d) 4

Solution:

$$\text{Average selling price} = \frac{1.5\times10^8}{15{,}000} = \frac{1.5}{1.5}\times\frac{10^8}{10^4} = 10^4$$

$\therefore$ Only for product C the selling price is more than the average $= \frac{7}{5}\times10^4 > 10^4$

Choice (A)

What is the profit (in Rs.) obtained from the sales of product A?

- a) 800
- b) 900
- c) 500
- d) 250

Solution:

Selling price (in ₹) of product A = $\dfrac{\frac{5}{100} \times 1.5 \times 10^8}{\frac{10}{100} \times 1.5 \times 10^4}$

$= \frac{1}{2} \times 10^4 = \frac{1}{2} \times 10{,}000 = 5{,}000$

$\therefore$ The profit on each item of product A = Rs.900. Choice (B)

What is the profit percentage on product C?

- a) 78%
- b) 260%
- c) 268%
- d) 60%

Solution:

$$\text{Selling price (in. ₹) of product C} = \frac{\frac{35}{100} \times 1.5 \times 10^8}{\frac{25}{100} \times 1.5 \times 10^4}$$

$$= \frac{7}{5} \times 10^4 = 12{,}000$$

Profit (in ₹) = 12,000 – 3800 = 10,200

$$\text{Profit} = \frac{10200}{3800} \times 100 \simeq 268\%.$$

Choice (C)

For which product is the profit percentage, the highest?

- a) A
- b) C
- c) B
- d) D

Solution:

The percentage profit is maximum for product C = 268%. Choice (B)

What is the total profit percentage of the company?

- a) 81%
- b) 160%
- c) 121%
- d) 181%

Solution:

Total revenue = 1.5×10^8

Total manufacturing cost = $1500 \times 4100 + 4500 \times 4000 + 3750 \times 3800 + 3000 \times 3400 + 2750 \times 3200$

$= 1000\,(150 \times 41 + 500 \times 40 + 375 \times 38 + 3 \times 34 + 225 \times 32)$

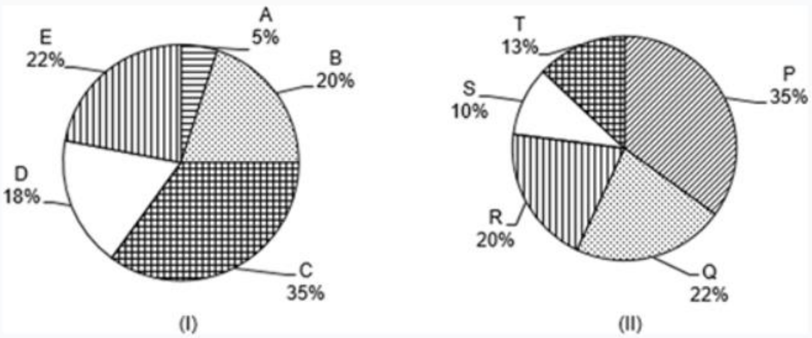
$= 1000\,(6150 + 20000 + 14250 + 10200 + 7200)$

$= 1000 \times 57400 = 5.74 \times 10^7$

Required profit percentage

$= \dfrac{1.5 \times 10^8 - 5.74 \times 10^7}{5.74 \times 10^7} \times 100 \approx 160\%.$ Choice (B)

Directions for questions 11 to 15: These questions are based on the following pie charts.



Pie chart (I) gives the percentage break-up of people participating in the discussions on various topics pertaining to Earth and its resources. The various topics are mentioned below:

A → Environmental degradation

B → Conservation of water

C → Adverse effects of global warming

D → Climatic variation

E → Depleting global oil reserves

Pie chart II gives the percentage distribution of the participating people, based on the country they come from.

P → United States of America (USA)

Q → United Kingdom (UK)

R → United Arab Emirates (UAE)

S → India

T → China

The number of persons participating in the discussion on the topic “Depleting global oil reserves in the world” is what percentage of those from the United Arab Emirates?

- a) 80%
- b) 90%
- c) 100%
- d) 110%

Solution:

From the information given in the pie chart, the answer is $\frac{22}{20} \times 100 = 110\%$.

Choice (D)

By what percentage is the number of people participating in the discussion on the topic "Environmental degradation" less than the number of people from the United Kingdom?

- a) $78\frac{2}{11}$
- b) $77\frac{3}{11}$
- c) 75
- d) $74\frac{1}{11}$

Solution:

From the pie chart, the percentage of people attending the discussion on "Environmental degradation" is 5% and the percentage of people attending the seminar from the United Kingdom is 22%

$\therefore \frac{17}{22} \times 100 = \frac{17}{11} \times 50 = \frac{850}{11} = 77\frac{3}{11}\%$ Choice (B)

What is the ratio of the number of people participating in the discussion on the topic "Adverse effects of global warming" to the number of people from India?

- a) 2 : 7
- b) 3 : 5
- c) 7 : 2
- d) 5 : 3

Solution:

The percentage of people attending the discussion on "Detrimental effects of Global warning" is 35% whereas the percentage of people from India is 10%

$\therefore$ The required ratio = 35 : 10 = 7 : 2. Choice (C)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

Had the number of people from China been 70 more, the percentage of people from China would have been 20. Find the total number of people who have participated in the discussions.

Solution:

From the pie chart, the percentage of people attending the seminar from China is 13%.
It is given that if there were 70 more people from China, then the percentage of people from China would have been 20%.
Let the total number of persons who attended the seminar initially be 100 K.
Similarly, the number of persons from China = 13 K.

Now, $\frac{13K+70}{100K+70} \times 100 = 20$

$\Rightarrow 5(13\text{ K} + 70) – 100\text{ K} + 70$

$\Rightarrow 35\text{ K} = 280$

$\Rightarrow \text{K} = 8.$

Therefore, there were initially 100 K or (100) 8 persons who attended the seminar.

Ans : (800)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

If the total number of persons participating in the discussions is 1500, then at most how many persons participating in the discussion on the topic “Conservation of water” are from China?

Solution:

Given:
The total number of persons attending the seminar = 1500.

$\Rightarrow$ The total number of persons from China = $\frac{13}{100} \times 1500 = 195.$

The number of persons attending the topic ‘B’ is $\frac{20}{100} \times 1500 = 300.$

$\therefore$ The maximum number of persons attending the topic ‘B’ from China is 195.

Ans : (195)